2:

Solution: No, consider $f : \mathbb{R} \to \mathbb{R}$ by $x \mapsto 0$. Take $A = \mathbb{R}$ and $x = 0$ so that x is a limit point of A but $f(A) = \{0\}$ which has no limit points since it is a singleton set: every neighborhood of 0 meets $f(A)$ only in 0. Thus $f(x) = 0$ is not a limit point of $f(A)$.

4:

Solution:

Proof. Without loss of generality we will show that $f : X \to X \times Y$ for $x \mapsto (x, y_0)$ is an embedding. It is clear that f is continuous since the sets $U \times V$ for $U \subseteq X$ open and $V \subseteq Y$ open form a basis for the product topology and

$$f^{-1}(U \times V) = \{x \mid (x, y_0) \in U \times V\} = \{x \in U \mid y_0 \in V\}$$

which is U if $y_0 \in V$ and empty otherwise. Both sets are open in X and hence f is continuous. It is also obvious that f is injective as if $f(x_1) = f(x_2)$ then

$$(x_1, y_0) = (x_2, y_0) \Rightarrow x_1 = x_2.$$

Now to show that f^{-1} is continuous we first notice that $f(X) = X \times \{y_0\}$, and the projection $\pi_X(f(x)) = x$. So, the inverse $f(X) \to X$ is the restriction of π_X to the subspace $f(X)$, which is continuous.
An almost identical argument exists for $g : Y \to X \times Y$. □

8:

Solution:

(a) *Proof.* We first define

$$h : X \to Y \times Y, \quad x \mapsto (f(x), g(x))$$

and

$$C = \{(a, b) \in Y \times Y \mid a \leq b\} \subseteq Y \times Y$$

so that

$$h^{-1}(C) = \{x \mid f(x) \leq g(x)\}$$

We now show that C is closed in $Y \times Y$. Take $(a, b) \in (Y \times Y) \setminus C$, so $a > b$. We find open sets $U \ni a$, $V \ni b$ with $U \times V \subseteq (Y \times Y) \setminus C$.

Case 1: there exists c with $b < c < a$. Take $U = (c, \infty)$ and $V = (-\infty, c)$. Then $a \in U$, $b \in V$, and for any $a' \in U$, $b' \in V$ we have $a' > c > b'$, so $(a', b') \notin C$.

Case 2: there is no such c. Take $U = (b, \infty)$ and $V = (-\infty, a)$, so $a \in U$ and $b \in V$. For $a' \in U$ and $b' \in V$ we have $a' > b$ and $b' < a$. Since nothing lies strictly between b and a, this gives $a' \geq a$ and $b' \leq b$, so

$$a' \geq a > b \geq b'$$

and thus $(a', b') \notin C$.

In both cases $U \times V$ is an open neighborhood of (a, b) contained in $(Y \times Y) \setminus C$. So $(Y \times Y) \setminus C$ is open and C is closed in $Y \times Y$. Since f and g are continuous, h is also continuous and thus $h^{-1}(C)$ is closed in X. □

(b) *Proof.* Using (a) (and (a) with the roles of f and g swapped), the sets

$$A = \{x \mid f(x) \leq g(x)\} \quad \text{and} \quad B = \{x \mid f(x) \geq g(x)\}$$

are closed in X. Let $m(x) = \min\{f(x), g(x)\}$. Since the order is total, every x satisfies $f(x) < g(x)$, $f(x) > g(x)$, or $f(x) = g(x)$, so $X = A \cup B$. On A we have $m(x) = f(x)$ and on B we have $m(x) = g(x)$, so $m|_A = f|_A$ and $m|_B = g|_B$ are continuous. If $x \in A \cap B$ then $f(x) = g(x)$, so the two formulas agree there. Thus, by the pasting lemma, m is continuous. □

9:

Solution:

(a) *Proof.* Let $\{A_\alpha\}$ be finite and each A_α closed. Then,

$$X = \bigcup_\alpha A_\alpha$$

and $f|_{A_\alpha}$ is continuous and for $i \neq j f|_{A_{\alpha_i}} = f|_{A_{\alpha_j}}$ for all $x \in A_{\alpha_i} \cap A_{\alpha_j}$. Thus, by the pasting lemma (for finitely many closed sets, by induction from the two-set version) we have that f is continuous by letting $f = f|_{A_{\alpha_i}}$ if $x \in A_{\alpha_i}$. □

(b) Let $X = [0,1]$, $A_\alpha = \left[\frac{1}{\alpha}, 1\right] \cup \{0\}$ for $\alpha \in \mathbb{N}$, and define $f : [0,1] \to \mathbb{R}$ by $f(0) = 0$ and $f(x) = 1/x$ for $x \neq 0$. Clearly

$$X = \bigcup_\alpha A_\alpha$$

and each A_α is closed, being a union of two closed sets. Also $f|_{A_\alpha}$ is continuous since $f(x) = 1/x$ is clearly continuous on $\left[\frac{1}{\alpha}, 1\right]$, and 0 is an isolated point of A_α. As well, f is not continuous at 0 since for $x_n = 1/n, x_n \to 0$ but $f(x_n) \to \infty$ thus $f(x_n)$ does not converge to 0. Thus, each A_α is closed and there are only a countable A_α (seen by the map $n \mapsto A_n$ for $n \in \mathbb{N}$) but f is not continuous.

(c) *Proof.* **Lemma.** If $\{D_\alpha\}$ is a locally finite family of closed sets in X, then $\bigcup_\alpha D_\alpha$ is closed.

Proof of Lemma. Let $x \notin \bigcup D_\alpha$. By local finiteness there is an open $U \ni x$ meeting only finitely many of the sets, say $D_1, \ldots, D_k$ ($k = 0$ is allowed). Let $V = U \setminus (D_1 \cup \cdots \cup D_k)$, which is open since a finite union of closed sets is closed, and $x \in V$ since x lies in no D_α. Then V misses every D_α: the sets other than $D_1, \ldots, D_k$ miss U, and $D_1, \ldots, D_k$ were removed. So every point of the complement has an open neighborhood inside the complement, and $\bigcup D_\alpha$ is closed.

Now assume $\{A_\alpha\}$ is locally finite and each A_α is closed. Let $C \subseteq Y$ be closed and set

$$B_\alpha = (f|_{A_\alpha})^{-1}(C) = A_\alpha \cap f^{-1}(C)$$

Since $X = \bigcup A_\alpha$ we know $f^{-1}(C) = \bigcup B_\alpha$ where B_α is closed in A_α because $f|_{A_\alpha}$ is continuous. It follows that B_α is closed in X since A_α is closed in X. Since $B_\alpha \subset A_\alpha$, any neighborhood that meets only finitely many A_α meets only finitely many B_α, thus making $\{B_\alpha\}$ locally finite. Thus, by the Lemma, $f^{-1}(C) = \bigcup B_\alpha$ is closed. Since C was an arbitrary closed set, f is continuous. □

2:

Solution:

Proof. Case 1: Assume both topologies have the box topology. Then, a basis for the box topology on X is

$$\{\Pi U_\alpha \mid U_\alpha \text{ open in } X_\alpha\}$$

thus, using $Y \cap \Pi U_\alpha = \Pi(A_\alpha \cap U_\alpha)$ (a point lies in both sides iff each coordinate lies in A_α and in U_α), a basis for the subspace topology on Y is

$$\beta_1 = \{\Pi(A_\alpha \cap U_\alpha) \mid U_\alpha \text{ open in } X_\alpha\}$$

A basis of the box topology on Y is

$$\{\Pi V_\alpha \mid V_\alpha \text{ open in } A_\alpha\}$$

but each $V_\alpha = A_\alpha \cap U_\alpha$ for U_α open in X_α. Thus,

$$\beta_2 = \{\Pi(A_\alpha \cap U_\alpha) \mid U_\alpha \text{ open in } X_\alpha\}$$

and $\beta_1 = \beta_2$ so their topologies are the same

Case 2: Assume both topologies are the product topologies which have basis

$$\{\Pi U_\alpha \mid U_\alpha \text{ open in } X_\alpha \text{ and } U_\alpha = X_\alpha \text{ for all but finitely many } \alpha\}$$

Now, consider a basis element of the subspace topology $Y \cap \Pi U_\alpha = \Pi(A_\alpha \cap U_\alpha)$. But, each $A_\alpha \cap U_\alpha$ is open in A_α, and for the α such that $U_\alpha = X_\alpha$, $A_\alpha \cap U_\alpha = A_\alpha \cap X_\alpha = A_\alpha$. Thus, this is a basis element of the product topology on Y. Take a basis element of the product topology on Y, ΠV_α where V_α is open in A_α and $V_\alpha \neq A_\alpha$ for finitely many α. For each of the finitely many $\alpha_1, \dots, \alpha_k$ such that $V_\alpha \neq A_\alpha$ choose U_α open in X_α with $V_\alpha = A_\alpha \cap U_\alpha$ and $U_\alpha = X_\alpha$ for all other α. Then ΠU_α is a basis element of the product topology on X, since $U_\alpha = X_\alpha$ for all but finitely many α. Also,

$$Y \cap \Pi U_\alpha = \Pi(A_\alpha \cap U_\alpha) = \Pi V_\alpha$$

for $\alpha \in \{\alpha_1, \dots, \alpha_k\}$. This is true by choice of U_α and for all other α, $A_\alpha \cap X_\alpha = A_\alpha = V_\alpha$. Thus, ΠV_α is open in the subspace topology and each topology contains each other's basis elements, and thus are equal. □

6:

Solution:

Proof. ($\Rightarrow$) Let U be a neighborhood of $\pi_\alpha(x)$ in X_α. Then $\pi_\alpha^{-1}(U)$ is open in the product by being a subbasis element, and it contains x. Since $x_i \to x$, we have that $x_i \in \pi_\alpha^{-1}(U)$ for i large, so $\pi_\alpha(x_i) \in U$ also for i large.
($\Leftarrow$) Let ΠU_α be a basis element containing x with $U_\alpha = X_\alpha$ for all α outside a finite set $\{\alpha_1, \dots, \alpha_k\}$. Then for each j, $\pi_{\alpha_j}(x_i) \to \pi_{\alpha_j}(x)$ and U_{α_j} is a neighborhood of $\pi_{\alpha_j}(x)$, so there is N_j with $\pi_{\alpha_j}(x_i) \in U_{\alpha_j}$ for $i \geq N_j$. We take $N = \max\{N_1, \dots, N_k\}$. For $i \geq N$ we have $\pi_\alpha(x_i) \in U_\alpha$ for **every** α (for $\alpha \in \{\alpha_1, \dots, \alpha_k\}$ by the choice of N, and trivially otherwise since $U_\alpha = X_\alpha$), so $x_i \in \Pi U_\alpha$. Hence $x_i \to x$. □

In the box topology on $\mathbb{R}^\omega$ take $x_i = \left(\frac{1}{i}, \frac{1}{i}, \dots\right)$ and $x = (0, 0, \dots)$. Then $\pi_n(x_i) = \frac{1}{i} \to 0$ but $U = \Pi\left(-\frac{1}{n}, \frac{1}{n}\right)$ is a box neighborhood of 0 and $x_i \in U$ implies $\frac{1}{i} < \frac{1}{n}$ for all n, which is impossible (take

$n = i$). So no x_i lies in U and $x_i \not\to x$ in the box topology. Thus the fact is **false** for the box topology, although ($\Rightarrow$) still holds there since the projections are still continuous.

7:

Solution: The closure of $\mathbb{R}^\infty$ is $\mathbb{R}^\omega$ in the product topology and $\mathbb{R}^\infty$ in the box topology.

Proof. Let $x = (x_1, x_2, \dots) \in \mathbb{R}^\omega$ and let ΠU_n be a basis element containing x, with $U_n = \mathbb{R}$ for $n > N$. Take $y = (x_1, \dots, x_N, 0, 0, \dots)$ so $y \in \mathbb{R}^\infty$, and $y \in U_n$ for all n (for $n \leq N, y_n = x_n$ and for $n > N$ since $U_n = \mathbb{R}$). So, every neighborhood of x meets $\mathbb{R}^\infty$, and the closure of $\mathbb{R}^\infty$ is all of $\mathbb{R}^\omega$.

We now show $\mathbb{R}^\infty$ is closed by showing its complement is open. Consider $x \notin \mathbb{R}^\infty$ so $x_n \neq 0$ for infinitely many n. Define

$$U_n = \begin{cases} \left(x_n - \frac{|x_n|}{2}, x_n + \frac{|x_n|}{2}\right) & \text{if } x_n \neq 0 \\ \mathbb{R} & \text{if } x_n = 0 \end{cases}$$

Then ΠU_n is a box-neighborhood of x. Any y in it has $y_n \neq 0$ whenever $x_n \neq 0$ (since $0 \notin U_n$), which is infinitely many n, so $y \notin \mathbb{R}^\infty$. Hence the complement is open so $\mathbb{R}^\infty$ is closed and thus it is its own closure. □

[0] LaTeX code for this document can be found on github here